

SAT Math · Phase 3 Final Review

1

★ Mark for Review

ABC

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

A. 171**B.** 168**C.** 165**D.** 45

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ABC

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

$$\begin{aligned}x^2 + y^2 &= 36 \\ y &= mx + \frac{b}{4}\end{aligned}$$

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

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ABC

A biologist mixed a solution that is 0.3% sodium chloride by mass with a solution that is 0.15% sodium chloride by mass to obtain a new solution, which has a mass of 80 grams and contains 0.21 grams of sodium chloride. How many grams of 0.3% sodium chloride solution did the biologist use?

A. 0.14**B.** 20**C.** 60**D.** 79.86

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ABC

The function v is defined by the equation

$$v(x) = \frac{x^2 + bx + c}{(x + 5)(x - 19)},$$

where b and c are constants. In the xy -plane, the graph of $y = v(x)$ passes through the points $(0, \frac{66}{95})$ and $(11, 0)$. If $v(q) = 0$, which of the following could be the value of q ?

A. -6 **B.** -5 **C.** 5 **D.** 6

5

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ABC

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b ?

A. 47

B. 4

C. $\frac{47}{45}$ D. $\frac{4}{45}$

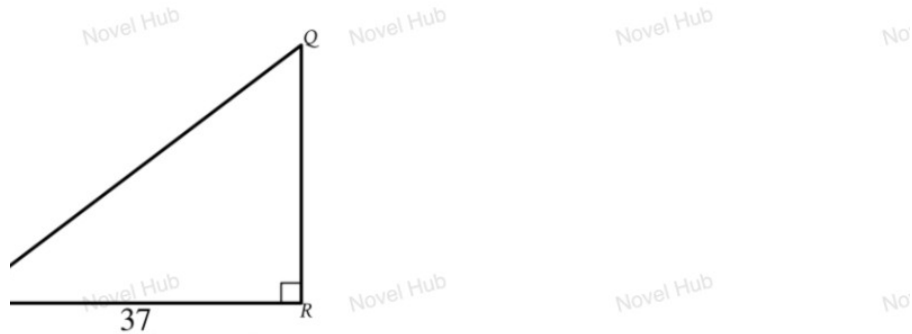
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ABC

Question 20

< :



Note: Figure not drawn to scale.

As shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A. $37 \cos Q$

B. $37 \sin Q$

C. $\frac{37}{\cos Q}$

D. $\frac{37}{\sin Q}$

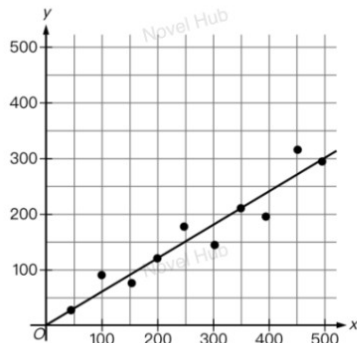
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ABC

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown.

relationship between the number of tomato seeds planted, x , and the number of tomato seeds that germinated, y , for each of the gardeners. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

- (A) The number of tomato seeds planted is predicted to increase by 60 seeds every 100 days.

Which of the following is the best interpretation of the slope of the line of best fit in this context?

A. The number of tomato seeds planted is predicted to increase by 60 seeds every 100 days.

B. The number of tomato seeds planted is predicted to increase by 300 seeds every 100 days.

C. The number of tomato seeds that germinate is predicted to increase by 60 seeds for every additional 100 tomato seeds that are planted.

D. The number of tomato seeds that germinate is predicted to increase by 300 seeds for every additional 100 tomato seeds that are planted.

8

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ABC

A rectangular banner has an area of 3,200 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

A. 3,220**B.** 3,240**C.** 3,840**D.** 4,608

The mass of object A is 444% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

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ABC

In 2013, Chloe earned 11% more than in 2012, and in 2014 Chloe earned 6% more than in 2013. If Chloe earned y times as much in 2012 as in 2014, which of the following is closest to the value of y ?

A. 0.5455**B.** 0.6600**C.** 0.8499**D.** 1.1766

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ABC

Point F lies on a unit circle in the xy -plane and has coordinates $(1,0)$. Point G is the center of the circle and has coordinates $(0,0)$. Point H also lies on the circle and has coordinates $(-1,y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

A. $\frac{37\pi}{2}$

B. $\frac{39\pi}{2}$

C. 34π

D. 35π